Analyzing Smart City Bike Sharing Data Using Power BI

1. Project Overview and Objective

This project focuses on cleaning, transforming, modelling and analysing Smart City Bike Sharing data using Power BI. The project aims to understand bike station availability, station capacity, city-wise performance and operational efficiency through interactive dashboards and data visualizations.

The main objective is to convert raw bike-sharing data into meaningful insights that help identify operational issues, compare city performance and support better bike-sharing resource management.

2. Data Sources

- **Dataset Source:** https://github.com/GeethaGunasekaran1/Project-Dataset/raw/refs/heads/main/bike-stations-sharing-data.xlsx
- **Dataset:** Bike Stations Sharing Data (Excel)
- **Domain:** Transportation Analytics

3. Problem Statement

The bike-sharing system generates a large amount of station-level data, including bike availability, station capacity, station status, and location details. Analysing this data manually makes it difficult to understand bike distribution, monitor station performance, and identify availability patterns across different cities.

This project focuses on transforming raw bike-sharing data into meaningful insights using Power BI. It aims to analyse bike availability, compare station capacities, monitor station status, identify trends, and support better decision-making for efficient bike-sharing system management.

4. Attribute (Column / Feature) Details

Attribute Name	Data Type	Description
number	Integer	Unique station identifier
name	Text	Bike station name
address	Text	Station address
position	Text	Geographic coordinates
banking	Boolean	Banking facility availability
bonus	Boolean	Bonus facility indicator
status	Text	Station operational status
Contract Name	Text	City or contract name
Bike Stands	Integer	Total station capacity

Available Bike Stands	Integer	Available empty stands
Available Bikes	Integer	Number of available bikes
Last Update	Date/Time	Last recorded station update

5. Tools and Technologies

- **Microsoft Excel:** Source dataset inspection and data handling.
- **Power BI Desktop:** Data modelling, DAX calculations and dashboard development.
- **Power Query:** Data cleaning and transformation.
- **DAX:** Calculated measures and statistical analysis.
- **Power BI Visualizations:** Interactive charts, KPI cards, maps and slicers.

6. Data Pre-Processing (Power Query)

The following cleaning and transformation operations were performed:

1. Imported the Excel dataset into Power BI.
2. Inspected column names, data types and data quality.
3. Checked missing values; no missing values were identified.
4. Checked duplicate records; no completely duplicated rows were identified.
5. Applied Trim and Clean operations to text columns.
6. Renamed columns for better readability.
7. Split the geographic position column into Latitude and Longitude.
8. Converted numeric columns into appropriate whole-number data types.
9. Converted the Last Update column into Date/Time.
10. Created a Capacity_Check column to identify inconsistencies in station capacity.
11. New column added : Available Bikes + Available Bike Stands = Bike Stands

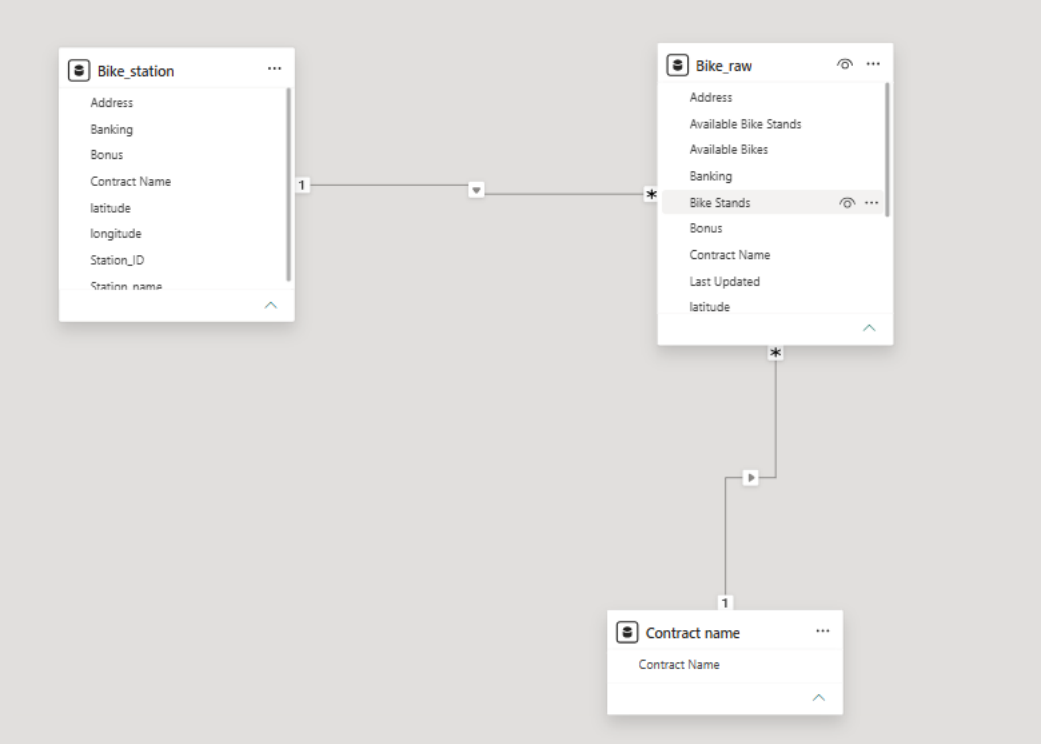
A total of 1,283 records were identified as capacity mismatches. These records were retained for further investigation rather than deleted.

Data Modelling:

The dataset was organized into a simple star-schema structure using:

- Bike_raw – Main station snapshot records.
- Bike_Station – Station information.
- Contract name – City information.

Relationships were established using one-to-many cardinality and single-direction filtering.



7. Data Modelling and DAX (Power BI)

Data Model

The dataset was organized into separate tables to improve data organization and analysis.

The following tables were used:

- **Bike_raw:** Main table containing bike station information, availability, capacity and station status.
- **Bike_station:** Dimension table containing station details such as station ID, station name, address, latitude and longitude.
- **Contract name:** Dimension table containing city/contract information.

Relationships were established using Station_ID and Contract Name with one-to-many cardinality and single-direction filtering.

DAX Measures Created

The following DAX measures were implemented in Power BI:

DAX Measure	Description
Total stations	Calculates the total number of distinct bike stations.
Available bikes	Calculates the number of available bikes.
Available empty stands	Calculates the number of available empty stands.
Bike occupying rate	Calculates the bike availability rate relative to total bike capacity.

Open station	Calculates the number of open station records.
Closed station	Calculates the number of closed station records.
Average	Calculates the average value used for station-wise analysis.

Total Stations

Total stations = DISTINCTCOUNT(Bike_raw[Station_ID])

2. Total Bike Capacity

Total bike capacity = SUM(Bike_raw[Total_capacity])

3. Available Bikes

Available bikes = SUM(Bike_raw[Available Bikes])

4. Available Empty Stands

Available empty stands = SUM(Bike_raw[Available Bike Stands])

5. Bike Occupying Rate

Bike occupying rate = DIVIDE([Available bikes], [Total bike capacity], 0)

6. Open Stations

Open station = CALCULATE(DISTINCTCOUNT(Bike_raw[Station_ID]), Bike_raw[Status] = "OPEN")

7. Closed Stations

Closed station =CALCULATE(DISTINCTCOUNT(Bike_raw[Station_ID]), Bike_raw[Status] = "CLOSED”)

8. Average Available Bikes

Average = AVERAGE(Bike_raw[Available Bikes])

9. Average Empty Stands

Average empty stands = AVERAGE(Bike_raw[Available Bike Stands])

10. Average Station Capacity

Average station capacity = AVERAGE(Bike_raw[Total_capacity])

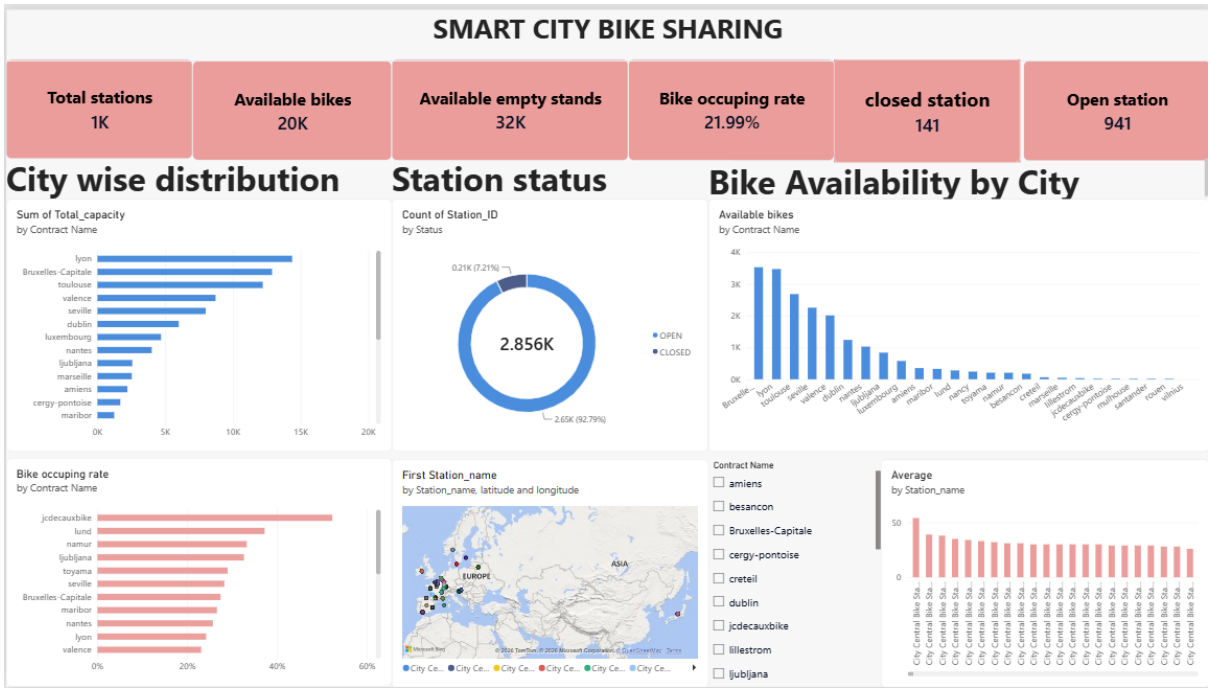
8. Analysis and Visualizations (Power BI)

An interactive Power BI dashboard was developed to analyse bike-sharing station performance and availability across different cities.

Dashboard Features and Visualizations

Visualization	Chart Type	Purpose
Total Stations	Card	Displays the total number of bike stations.

Available Bikes	Card	Displays available bike information.
Available Empty Stands	Card	Displays empty stand information.
Bike Occupying Rate	Card	Displays bike occupancy percentage.
Open Station	Card	Displays open station records.
Closed Station	Card	Displays closed station records.
Total Capacity by City	Bar Chart	Compares bike capacity across cities.
Station Status	Donut Chart	Shows the distribution of open and closed station records.
Bike Availability by City	Column Chart	Compares available bikes across cities.
Bike Occupying Rate by City	Bar Chart	Compares bike occupancy rates across cities.
Station-wise Average	Combo Chart	Compares average values across individual stations.
Station Locations	Map	Displays bike station locations using geographical coordinates



9. Insights & Conclusions

The dashboard provides insights into bike station distribution, availability and operational performance.

A. Descriptive Analysis

- Analysed the total number of bike stations across different cities.
- Examined available bikes and empty stands.
- Evaluated open and closed station records.
- Calculated bike occupancy rates.
- Compared average values across individual stations.

B. Diagnostic Analysis

- Compared bike availability across different cities to identify variations.
- Examined station status distribution to understand operational availability.
- Analysed bike occupancy rates to identify differences in station capacity utilization.
- Used geographical visualizations to understand the distribution of bike stations.

C. Predictive Analysis

- Identify stations with consistently high or low occupancy to support better capacity management.
- Identify stations that may experience low bike availability in the future.

D. Prescriptive Analysis and Recommendations

- Improve bike redistribution between stations based on availability.
- Monitor stations with low bike availability.
- Investigate frequently closed stations.
- Optimize bike capacity based on station requirements.
- Monitor station occupancy to improve resource utilization.
- Use geographical station information for better operational planning.

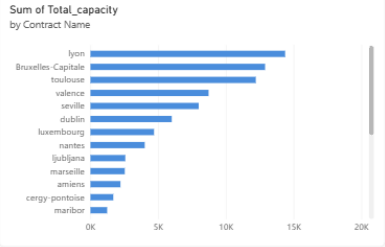
10. Conclusions

The dashboard provides a consolidated view of bike station performance, city-wise bike availability, station status and capacity utilization.

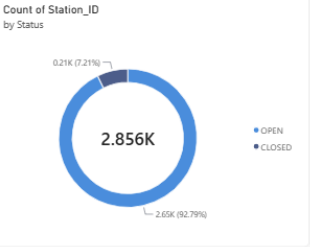
SMART CITY BIKE SHARING

Total stations	Available bikes	Available empty stands	Bike occuping rate	closed station	Open station
1K	20K	32K	21.99%	141	941

City wise distribution



Station status



Bike Availability by City

